

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 5. Conditional Statements

Lecture: 44. Challenge - Weekday Name

Section 1: Lecture Summary

This lecture covers three programming challenges that involve using nested if-elif-else structures in Python. The challenges are closely related and focus on mapping numeric inputs to corresponding names or words. The first challenge requires converting a day number (0 to 6) into its weekday name. The second challenge involves converting a month number (typically 0 to 11) into the month name. The third challenge is about converting a single digit (0 to 9) into its word equivalent. Each problem is explained conceptually with calendar examples, and the instructor encourages learners to implement the solutions using nested if-elif ladders. Sample program structures are demonstrated for the day and month challenges, showing how to check multiple conditions and print the corresponding output. The lecture emphasizes practice and hands-on programming to reinforce these concepts.

Section 2: Key Concepts and Explanations

- Nested if-elif-else Structures:
 - Used to test multiple conditions sequentially.
 - Enables mapping of input values (numbers) to output values (names or words).
 - The structure checks each condition in order; if a condition is met, the corresponding block executes, and the rest are skipped.
 - An optional else clause handles invalid or unexpected inputs.
- Day Number to Name Mapping:
 - Weekdays can be numbered from 0 to 6, with Monday as 0.

- The program matches each number to its weekday name (e.g., 0 → Monday, 1 → Tuesday, ..., 6 → Sunday).

- Input: An integer representing the day number.

- Output: The corresponding day name or "Invalid day number" for out-of-range inputs.

- Month Number to Name Mapping:

- Months can be numbered starting from 0 (January) to 11 (December).

- The program maps these numbers to the month names.

- Input: An integer representing the month number.

- Output: The corresponding month name or "Invalid month number" for out-of-range inputs.

- Digit to Word Mapping:

- Single-digit integers from 0 to 9 are converted to their English words.

- Input: A single digit integer.

- Output: The English word equivalent of the digit or "Invalid digit" if the input is outside 0-9.

- Input Handling:

- The programs accept input from the user using `input()` and convert them to integers using `int()`.

- Validation is performed using nested if-elif-else to ensure the input is within the expected range.

- Importance of Practice:

- The instructor emphasizes practicing these patterns as they form the foundation for understanding conditional logic in Python.

Section 3: Example Code and Use Cases

1. Day Number to Day Name

```
dayNo = int(input("Enter a day number (0-6): "))

if dayNo == 0:
    print("Monday")
elif dayNo == 1:
    print("Tuesday")
elif dayNo == 2:
    print("Wednesday")
elif dayNo == 3:
    print("Thursday")
elif dayNo == 4:
    print("Friday")
elif dayNo == 5:
    print("Saturday")
elif dayNo == 6:
    print("Sunday")
else:
    print("Invalid day number")
```

This program takes a day number as input and prints the corresponding weekday name. If the number is out of range, it prints an invalid message.

2. Month Number to Month Name

```
monthNo = int(input("Enter a month number (0-11): "))

if monthNo == 0:
    print("January")
elif monthNo == 1:
    print("February")
elif monthNo == 2:
    print("March")
elif monthNo == 3:
    print("April")
elif monthNo == 4:
```

```
    print("May")
elif monthNo == 5:
    print("June")
elif monthNo == 6:
    print("July")
elif monthNo == 7:
    print("August")
elif monthNo == 8:
    print("September")
elif monthNo == 9:
    print("October")
elif monthNo == 10:
    print("November")
elif monthNo == 11:
    print("December")
else:
    print("Invalid month number")
```

This program maps a month number to the respective month name and handles invalid inputs.

3. Digit to Word

```
digit = int(input("Enter a single digit (0-9): "))

if digit == 0:
    print("Zero")
elif digit == 1:
    print("One")
elif digit == 2:
    print("Two")
elif digit == 3:
    print("Three")
elif digit == 4:
    print("Four")
elif digit == 5:
    print("Five")
elif digit == 6:
    print("Six")
elif digit == 7:
    print("Seven")
elif digit == 8:
```



```
    print("Eight")
elif digit == 9:
    print("Nine")
else:
    print("Invalid digit")
```

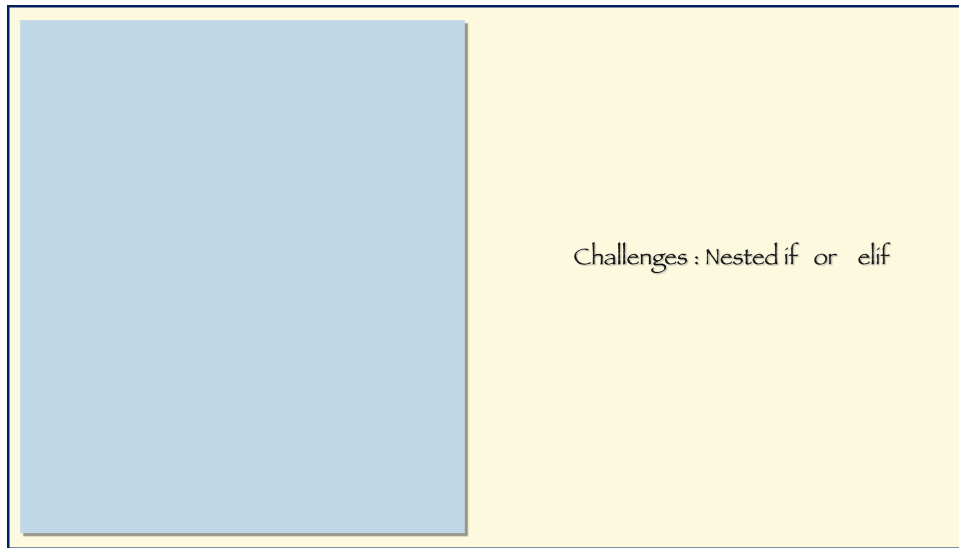
This code converts a digit input to its word form, or outputs an invalid message for numbers outside 0-9.

Section 4: Key Takeaways

- Nested if-elif-else statements allow checking multiple conditions and controlling program flow accordingly.
- Day, month, and digit name challenges demonstrate practical use cases for conditional ladders.
- Indexing days and months from zero is a common convention in programming.
- Validating input ensures programs behave as expected and provide meaningful error messages.
- Practice by writing and testing such conditional programs improves understanding and coding skills.
- Watching lectures alone is insufficient; active coding practice is essential to learning programming effectively.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Challenges : Nested if or elif

- Day No. to Name

Challenges : Nested if or elif

- Day No. to Name
- Month No. to Name

Challenges : Nested if or elif

- Day No. to Name
- Month No. to Name
- Digit in Words

- Day No. to Name

Day Number to Day Name
Challenge

Day Number to Day Name
Challenge

January

Day Number to Day Name

Challenge

January

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number to Day Name

Challenge

January

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Name

Mon

Tue

Wed

Thur

Fri

Sat

Sun

Day Number to Day Name

Challenge

January

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

01

Day Name

Mon	Tue	Wed	Thur	Fri	Sat	Sun
-----	-----	-----	------	-----	-----	-----

Day Number to Day Name

Challenge

January

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

012

Day Name

Mon	Tue	Wed	Thur	Fri	Sat	Sun
-----	-----	-----	------	-----	-----	-----

Day Number to Day Name
Challenge

January						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number 0 1 2 3 4 5 6

Day Name Mon Tue Wed Thur Fri Sat Sun

Day Number to Day Name
Challenge

January

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number

0123456

Day Name

Mon	Tue	Wed	Thur	Fri	Sat	Sun
-----	-----	-----	------	-----	-----	-----

dayNo = ?

Day Number to Day Name
Challenge

January

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number

0123456

Day Name

Mon	Tue	Wed	Thur	Fri	Sat	Sun
-----	-----	-----	------	-----	-----	-----

dayNo = 1 'Tuesday'

dayNo = ?

Day Number to Day Name Challenge

January						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number 0 1 2 3 4 5 6
Day Name Mon Tue Wed Thur Fri Sat Sun

dayNo = ?

dayNo = ?

if dayNo == 0:

Day Number to Day Name Challenge

January						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number 0 1 2 3 4 5 6
Day Name Mon Tue Wed Thur Fri Sat Sun

dayNo = ?

```
dayNo = ?  
  
if dayNo == 0:  
    print( 'Monday' )
```

Day Number to Day Name Challenge

January						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number 0 1 2 3 4 5 6
Day Name Mon Tue Wed Thur Fri Sat Sun

dayNo = ?

```
dayNo = ?  
  
if dayNo == 0:  
    print( 'Monday' )  
elif dayNo == 1:
```

Day Number to Day Name Challenge

January						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number 0 1 2 3 4 5 6
Day Name Mon Tue Wed Thur Fri Sat Sun

dayNo = ?

```
dayNo = ?  
  
if dayNo == 0:  
    print( 'Monday' )  
  
elif dayNo == 1 :  
    print( 'Tuesday' )
```

Day Number to Day Name Challenge

January						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number 0 1 2 3 4 5 6
Day Name Mon Tue Wed Thur Fri Sat Sun

dayNo = ?

```
dayNo = ?  
  
if dayNo == 0:  
    print( 'Monday' )  
  
elif dayNo == 1 :  
    print( 'Tuesday' )  
  
elif dayNo == 2 :
```

Day Number to Day Name Challenge

January						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number 0 1 2 3 4 5 6
Day Name Mon Tue Wed Thur Fri Sat Sun

dayNo = ?

```
dayNo = ?  
  
if dayNo == 0:  
    print( 'Monday' )  
elif dayNo == 1 :  
    print( 'Tuesday' )  
elif dayNo == 2 :  
    print( 'Wednesday' )  
elif dayNo == 3 :  
    print( 'Thursday' )  
elif dayNo == 4 :  
    print( 'Friday' )  
elif dayNo == 5 :  
    print( 'Saturday' )  
elif dayNo == 6 :  
    print( 'Sunday' )
```

Day Number to Day Name Challenge

January						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number 0 1 2 3 4 5 6
Day Name Mon Tue Wed Thur Fri Sat Sun

dayNo = ?

```
dayNo = ?  
  
if dayNo == 0:  
    print( 'Monday' )  
elif dayNo == 1 :  
    print( 'Tuesday' )  
elif dayNo == 2 :  
    print( 'Wednesday' )  
elif dayNo == 3 :  
    print( 'Thursday' )  
elif dayNo == 4 :  
    print( 'Friday' )  
elif dayNo == 5 :  
    print( 'Saturday' )  
elif dayNo == 6 :  
    print( 'Sunday' )  
else:
```

Day Number to Day Name Challenge

January						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number 0 1 2 3 4 5 6
Day Name Mon Tue Wed Thur Fri Sat Sun

dayNo = ?

```
dayNo = int( input( 'Enter a Day Number' ) )  
  
if dayNo == 0 :  
    print( 'Monday' )  
elif dayNo == 1 :  
    print( 'Tuesday' )  
elif dayNo == 2 :  
    print( 'Wednesday' )  
elif dayNo == 3 :  
    print( 'Thursday' )  
elif dayNo == 4 :  
    print( 'Friday' )  
elif dayNo == 5 :  
    print( 'Saturday' )  
elif dayNo == 6 :  
    print( 'Sunday' )  
else:  
    print( 'Invalid Day Number' )
```

Day Number to Day Name Challenge

January						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number 0 1 2 3 4 5 6
Day Name Mon Tue Wed Thur Fri Sat Sun

dayNo = ?

```
dayNo = int( input( 'Enter a Day Number' ))

if dayNo == 0:
    print( 'Monday' )
elif dayNo == 1:
    print( 'Tuesday' )
elif dayNo == 2:
    print( 'Wednesday' )
elif dayNo == 3:
    print( 'Thursday' )
elif dayNo == 4:
    print( 'Friday' )
elif dayNo == 5:
    print( 'Saturday' )
elif dayNo == 6:
    print( 'Sunday' )
else:
    print( 'Invalid Day Number' )
```

Day Number to Day Name
Challenge

January						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Day Number

0123456

Day Name

MonTueWedThurFriSatSun

dayNo = ?

- Month No. to Name

Month Number to Name
Challenge

Month Number to Name

Challenge

January SMTWTFSS 31 3 10 17 24	February SMTWTFSS 28 7 14 21 28	March SMTWTFSS 31 6 13 20 27	April SMTWTFSS 30 3 10 17 24
May SMTWTFSS 31 8 15 22 29	June SMTWTFSS 30 5 12 19 26	July SMTWTFSS 31 3 10 17 24	August SMTWTFSS 31 7 14 21 28
September SMTWTFSS 30 4 11 18 25	October SMTWTFSS 31 2 9 16 23	November SMTWTFSS 30 6 13 20 27	December SMTWTFSS 31 4 11 18 25

Month Number to Name

Challenge

January	February	March	April
May	June	July	August
September	October	November	December

Month Number to Name

Challenge

Month Name

January

February

March

April

May

June

July

August

September

October

November

December

Month Number to Name

Challenge

Month Name

0 January

1 February

2 March

3 April

4 May

5 June

6 July

7 August

8 September

9 October

10 November

11 December

Month Number to Name

Challenge

monthNo = ?

Month Name

0 January

1 February

2 March

3 April

4 May

5 June

6 July

7 August

8 September

9 October

10 November

11 December

Month Number to Name

Challenge

monthNo = 5

‘June’

Month Name

0 January

1 February

2 March

3 April

4 May

5 June

6 July

7 August

8 September

9 October

10 November

11 December

Month Number to Name
Challenge

monthNo = 8
'September'

Month Name	
0	January
1	February
2	March
3	April
4	May
5	June
6	July
7	August
8	September
9	October
10	November
11	December

monthNo = ?

Month Number to Name

Challenge

monthNo = ?

Month Name

0	January
1	February
2	March
3	April
4	May
5	June
6	July
7	August
8	September
9	October
10	November
11	December

monthNo = ?

if monthNo == 0 :

Month Number to Name

Challenge

monthNo = ?

Month Name

0	January
1	February
2	March
3	April
4	May
5	June
6	July
7	August
8	September
9	October
10	November
11	December

monthNo = ?

if monthNo == 0 :

print('January')

monthNo = ?

Month Number to Name

Challenge

monthNo = ?

Month Name

0 January

1 February

2 March

3 April

4 May

5 June

6 July

7 August

8 September

9 October

10 November

11 December

monthNo = ?

if monthNo == 0 :

print('January')

elif monthNo == 1 :

monthNo = ?

Month Number to Name

Challenge

monthNo = ?

Month Name

0 January

1 February

2 March

3 April

4 May

5 June

6 July

7 August

8 September

9 October

10 November

11 December

monthNo = ?

if monthNo == 0 :

print('January')

elif monthNo == 1 :

print('February')

Month Number to Name

Challenge

monthNo = ?

Month Name

0

January

1

February

2

March

3

April

4

May

5

June

6

July

7

August

8

September

9

October

10

November

11

December

monthNo = ?

if monthNo == 0 :

print('January')

elif monthNo == 1 :

print('February')

elif monthNo == 2 :

Month Number to Name

Challenge

monthNo = ?

Month Name

0

January

1

February

2

March

3

April

4

May

5

June

6

July

7

August

8

September

9

October

10

November

11

December

monthNo = ?

if monthNo == 0 :

print('January')

elif monthNo == 1 :

print('February')

elif monthNo == 2 :

print('March')

Month Number to Name

Challenge

monthNo = ?

	Month Name
0	January
1	February
2	March
3	April
4	May
5	June
6	July
7	August
8	September
9	October
10	November
11	December

- Digit in Words

Digit in Words
Challenge

Digit in Words
Challenge

Digits

Digit in Words
Challenge

Digits

0
1
2
3
4
5
6
7
8
9

Digit in Words

Challenge

Digits

0

1

2

3

4

5

6

7

8

9

Zero

Digit in Words

Challenge

Digits

0

1

2

3

4

5

6

7

8

9

Zero

One

Digit in Words

Challenge

Digits

0

Zero

1

One

2

Two

3

4

5

6

7

8

9

Digit in Words

Challenge

Digits

0

Zero

1

One

2

Two

3

Three

4

5

6

7

8

9

Digit in Words

Challenge

Digits

0

Zero

1

One

2

Two

3

Three

4

Four

5

6

7

8

9

Digit in Words

Challenge

Digits

0

Zero

1

One

2

Two

3

Three

4

Four

5

Five

6

7

8

9

Digit in Words

Challenge

Digits

0	Zero
1	One
2	Two
3	Three
4	Four
5	Five
6	Six
7	
8	
9	

Digit in Words

Challenge

Digits

0	Zero
1	One
2	Two
3	Three
4	Four
5	Five
6	Six
7	Seven
8	
9	

Digit in Words

Challenge

Digits

0	Zero
1	One
2	Two
3	Three
4	Four
5	Five
6	Six
7	Seven
8	Eight
9	

Digit in Words

Challenge

Digits

0	Zero
1	One
2	Two
3	Three
4	Four
5	Five
6	Six
7	Seven
8	Eight
9	Nine

Digit in Words
Challenge

digit = ?

Digits

0	Zero
1	One
2	Two
3	Three
4	Four
5	Five
6	Six
7	Seven
8	Eight
9	Nine

Digit in Words

Challenge

digit = 8

Digits

0

Zero

1

One

2

Two

3

Three

4

Four

5

Five

6

Six

7

Seven

8

Eight

9

Nine

Digit in Words

Challenge

digit = 8

'Eight'

Digits

0

Zero

1

One

2

Two

3

Three

4

Four

5

Five

6

Six

7

Seven

8

Eight

9

Nine

digit = 5

'Five'

Digit in Words

Challenge

Digits

0	Zero
1	One
2	Two
3	Three
4	Four
5	Five
6	Six
7	Seven
8	Eight
9	Nine

digit = ?

digit = ?

Digit in Words

Challenge

Digits

0	Zero
1	One
2	Two
3	Three
4	Four
5	Five
6	Six
7	Seven
8	Eight
9	Nine

digit = ?

if digit == 0:

digit = ?

Digit in Words

Challenge

Digits

0	Zero
1	One
2	Two
3	Three
4	Four
5	Five
6	Six
7	Seven
8	Eight
9	Nine

digit = ?

if digit == 0:

print('Zero')

digit = ?

Digit in Words

Challenge

Digits

0	Zero
1	One
2	Two
3	Three
4	Four
5	Five
6	Six
7	Seven
8	Eight
9	Nine

digit = ?

if digit == 0 :
 print('Zero')
elif digit == 1 :

digit = ?

Digit in Words

Challenge

Digits

0	Zero
1	One
2	Two
3	Three
4	Four
5	Five
6	Six
7	Seven
8	Eight
9	Nine

digit = ?

if digit == 0 :
 print('Zero')
elif digit == 1 :
 print('One')

digit = ?

Digit in Words

Challenge

Digits

0	Zero
1	One
2	Two
3	Three
4	Four
5	Five
6	Six
7	Seven
8	Eight
9	Nine

digit = ?

if digit == 0 :

print('Zero')

elif digit == 1 :

print('One')

elif digit == 2 :

Digit in Words

Challenge

digit = ?

Digits

0

Zero

1

One

2

Two

3

Three

4

Four

5

Five

6

Six

7

Seven

8

Eight

9

Nine

digit = ?

if digit == 0 :

print('Zero')

elif digit == 1 :

print('One')

elif digit == 2 :

print('Two')

Digit in Words

Challenge

digit = ?

Digits

0

Zero

1

One

2

Two

3

Three

4

Four

5

Five

6

Six

7

Seven

8

Eight

9

Nine

digit = ?

if digit == 0 :

print('Zero')

elif digit == 1 :

print('One')

elif digit == 2 :

print('Two')

.

.

.

.

.

.

.

Digit in Words

Challenge

digit = ?

Digits

0	Zero
1	One
2	Two
3	Three
4	Four
5	Five
6	Six
7	Seven
8	Eight
9	Nine